Affan Muhammad Shaikh

ffaanshake@gmail.com | linkedin.com/in/sil6428 | github.com/sil6428 | affan-shaikh.pages.dev

Education

Ontario Tech University Oshawa, ON

Bachelor of Information Technology (Honours) - Networking and IT Security *Sept. 2024 - Expected Apr. 2028*

- Coursework: Computer Security, Cryptography, Cloud Computing, Advanced Networking.

Experience

Co-Founder and Platform Developer May 2026 - Present

SSIK IT Consulting & Solutions - Student-Led Initiative *Ontario, Canada*

- Co-founded a student-led IT and cybersecurity initiative with one partner; define **service scopes**, **privacy research**, control-review methods, and client-ready recommendations.
- Built a private internal review platform for tracking organizations, evidence, findings, and review states with **access controls**, outbound-request safeguards, **audit logs**, and verified backups.

Head of Internal Technical Operations Jan. 2026 - Present

Arch Narrative - Developing Nonprofit *Oshawa, ON*

- Built and manage internal systems used by **200+** CASIC volunteers, students, staff, and organizational leaders.
- Administer **Google Workspace**, access, permissions, internal tools, repositories, and deployment workflows.

Projects

OTNow Canvas Companion | *Chrome Manifest V3, JavaScript, Canvas APIs*

- Designed and published a **Chrome Web Store** extension that groups Canvas deadlines, detects moved dates, schedules reminders, links course resources, and caches data for offline viewing.
- Restricted Canvas access to **2** allowlisted read-only endpoints and stored coursework locally in Chrome storage.
- Opt-in telemetry has recorded **450+** organized coursework items while non-participant data remains local.

Secure File Transfer | *Python, TLS, SHA-256, FastAPI*

- Built an authenticated **TLS** client/server and loopback-only browser workspace for certificate setup, account management, resumable uploads, recipient inboxes, and verified downloads.
- Implemented recipient-scoped storage, exact-offset resumption, two-sided **SHA-256** verification, tamper quarantine, throttling, and audit logging across upload and download paths.
- Benchmarked with **16 concurrent workers**, completing **2K** verified TLS transfers across **1K** unique files while returning all recipient inbox records within bounded protocol frames.

Cisco Networking Labs | *Cisco IOS, Packet Tracer, Wireshark*

- Built and verified multi-device routed and switched topologies using **IPv4/IPv6**, **VLANs**, **802.1Q** trunks, inter-VLAN routing, DHCP, STP, NAT, and OSPF/EIGRP fundamentals.
- Isolated connectivity and routing faults with **Cisco IOS** operational state, routing tables, ping, traceroute, and **Wireshark** captures; revalidated reachability after each correction.

P2P Messaging (Collaborative WIP) | *Python, FastAPI, TCP/IP, ChaCha20-Poly1305*

- Co-developing a local encrypted messenger using **Ed25519/X25519** device identities and **ChaCha20-Poly1305** messages, with signed peer cards and fingerprint verification.
- Added authenticated acknowledgements, persistent replay rejection, rate limits, and encrypted local history with **64 KiB** frame and **4 KiB** plaintext limits; documented missing forward secrecy and NAT traversal.

Technical Skills

Networking: IPv4/IPv6, subnetting, VLANs, 802.1Q, inter-VLAN routing, DHCP, DNS, NAT, STP, OSPF/EIGRP fundamentals, routing and switching troubleshooting
Security: TLS, SHA-256, file-integrity monitoring, access control, audit logging, input validation, threat modelling
Languages: Python, JavaScript, TypeScript, HTML/CSS
Systems: Linux, Windows Server, Cisco IOS, Wireshark, Packet Tracer, Git, Chrome extensions, Canvas APIs